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Academic Task : Project	**Course instructor: Robin Tyagi**
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Maximum Marks: 50	**Section: Q5E09**
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Declaration:

I declare that this Assignment is our individual work. I have not copied it from any other students' work or from any other source except where due acknowledgement is made explicitly in the text, nor has any part been written for me by any other person.

Evaluation Criterion: Rubrics on different parameters

Evaluator's Comments (For Instructor's use only)

General Observations	Suggestions for Improvement	Best part of assignment

Evaluator's Signature and Date:

Project Overview

This report documents an end-to-end data mining workflow developed in IBM SPSS Modeler for predicting customer churn in the telecommunications industry.

The primary business goal is to identify customers who are likely to churn (i.e., leave the service), enabling the telecom company to take proactive retention measures. A CHAID (Chi-squared Automatic Interaction Detection) decision tree model was built and trained on historical customer data.

1.1 Objectives

- Understand the structure and quality of the telecom customer dataset.
- Filter and prepare the data for modeling using known customer records.
- Build a CHAID decision tree to classify customers as churned or not churned.
- Evaluate the model and interpret the decision tree output.
- Follow the CRISP-DM lifecycle throughout the project.

2. CRISP-DM Framework

The CRISP-DM (Cross-Industry Standard Process for Data Mining) methodology was used to structure this project. This framework consists of six phases that guide a data mining project from conception to deployment. All six phases are visible in the SPSS Modeler project panel on the right side of the workspace.

Phase	Description	Status
Business Understanding	Define goals: predict customer churn to improve retention	Complete
Data Understanding	Explore the telecom dataset, review fields and data types	Complete
Data Preparation	Filter records where DATA_KNOWN = 'yes'; set field types	Complete
Modeling	Build CHAID decision tree with HAS_CHURNED as target	Complete
Evaluation	Review model accuracy, tree structure, and predictions	Complete
Deployment	Model nugget saved for scoring new/unknown data	Complete

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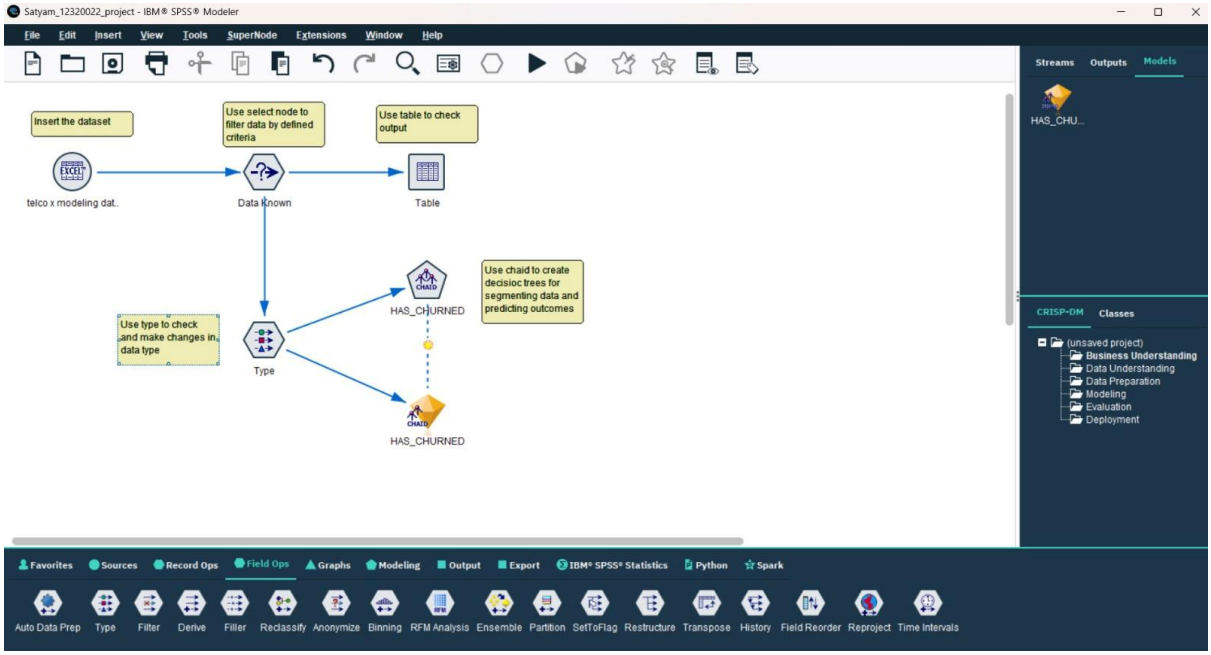
3. SPSS Modeler Stream Overview

The SPSS Modeler stream (canvas) contains the complete data flow pipeline from data ingestion to model output. The stream is composed of five key nodes connected in sequence, with annotations describing the purpose of each node.



3.1 Stream Pipeline

Step	Node	Node Type	Purpose
1	telco x modeling dat..	Excel Source Node	Imports the raw telecom customer dataset from Excel
2	Data Known	Select Node	Filters rows where DATA_KNOWN = 'yes'
3	Type	Type Node	Defines measurement levels and roles for each field
4	HAS_CHURNED (model)	CHAID Modeling Node	Trains the CHAID decision tree model
5	HAS_CHURNED (nugget)	CHAID Model Nugget	Applies the trained model to generate predictions
6	Table	Table Output Node	Displays the output table to verify data and results

The stream has two branches from the Type node:

- One branch leads to the CHAID modeling node which trains and generates the model nugget.
- The model nugget (golden diamond icon) is then connected downstream to apply the trained model to score records.

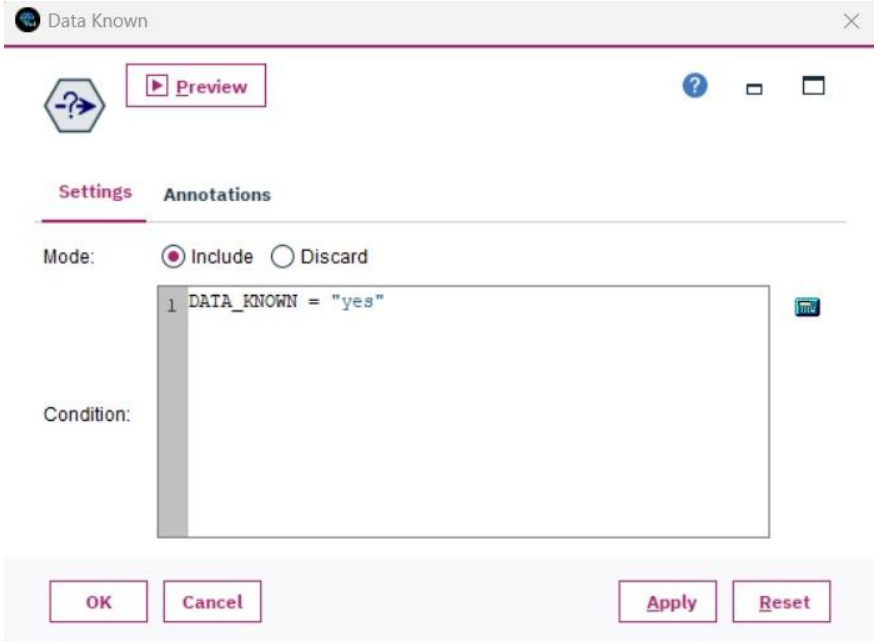
4. Detailed Node Descriptions

4.1 Data Source Node — telco x modeling dat..

The stream begins with an Excel source node that loads the raw telecom customer dataset. This file contains records with a wide range of customer attributes including demographics, tariff plan, usage metrics, handset type, and gadget-related information.

4.2 Select Node — Data Known

The Select node filters the incoming dataset to retain only records where the field DATA_KNOWN equals "yes". This step ensures that the model is trained only on customers whose churn status is already known (historical/labeled data).



4.3 Type Node — Type

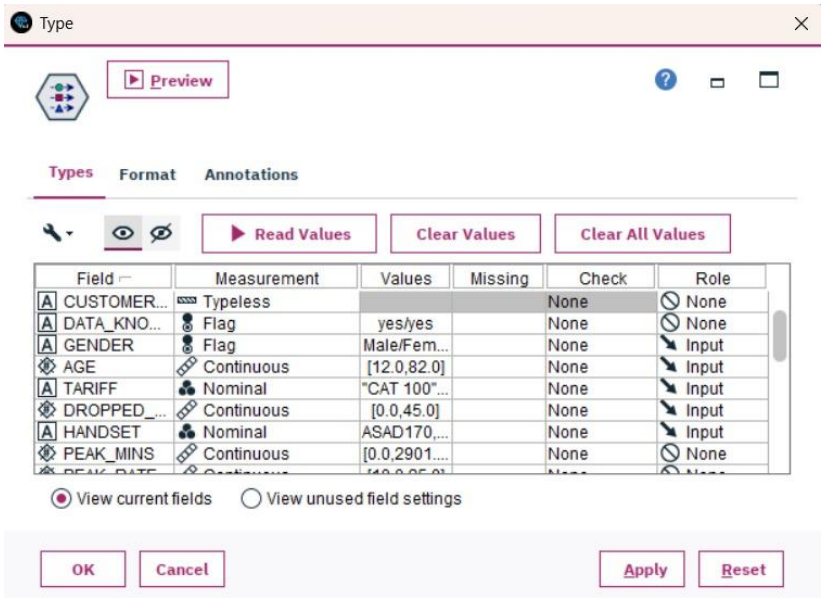
The Type node is one of the most important nodes in the stream. It defines the measurement level (data type) and the role (Input, Target, None) for each field in the dataset. This metadata is essential because SPSS Modeler uses it to correctly interpret variables during modeling.

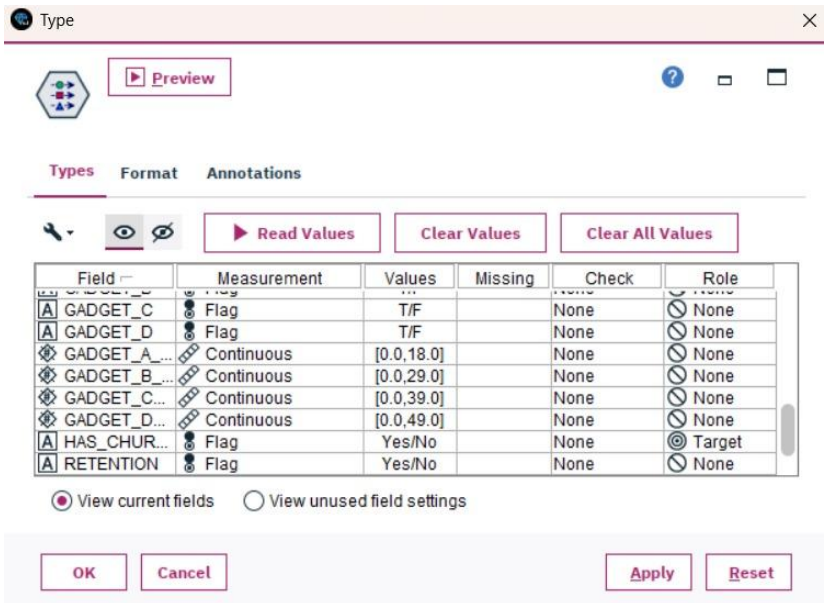
Field Type Configuration:

Field	Measurement	Values	Role
CUSTOMER_ID	Typeless	—	None (excluded)
DATA_KNOWN	Flag	yes / yes	None (not used in model)
GENDER	Flag	Male / Female	Input
AGE	Continuous	[12.0, 82.0]	Input
TARIFF	Nominal	CAT 100, etc.	Input
DROPPED_CALLS	Continuous	[0.0, 45.0]	Input

Field	Measurement	Values	Role
HANDSET	Nominal	ASAD170, etc.	Input
PEAK_MINS	Continuous	[0.0, 2901...]	None
PEAK_RATE	Continuous	[10.0, 95.0]	Input
HAS_CHURNED	Flag	T / F	Target

Note: Fields with Role set to "None" (such as CUSTOMER_ID and DATA_KNOWN) are excluded from the modeling process but remain in the data stream. The target variable HAS_CHURNED is set to be predicted by the model.



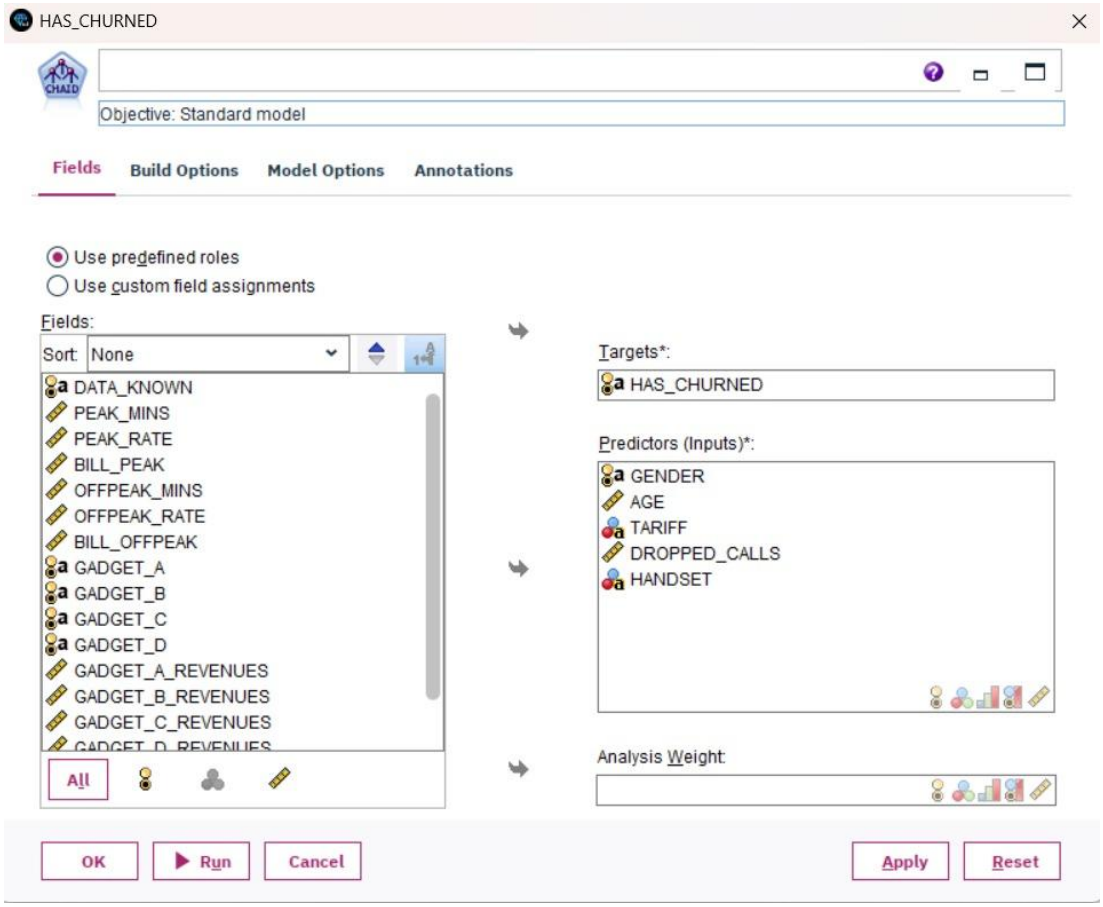


4.4 CHAID Modeling Node — HAS_CHURNED

The CHAID (Chi-squared Automatic Interaction Detection) algorithm is a decision tree method that splits the data based on the most statistically significant predictor at each node. It uses the chi-square test for categorical targets to determine which variable best splits the data.

Why CHAID?

- CHAID can produce multi-way splits (more than two branches per node), unlike binary trees such as CART.
- It is well-suited for segmenting customers based on categorical and continuous inputs.
- It provides interpretable decision rules which are easy to explain to business stakeholders.
- It handles both nominal and continuous predictor variables effectively.



4.5 CHAID Model Nugget — HAS_CHURNED

After the modeling node processes the data, it generates a Model Nugget — represented by the golden diamond icon in SPSS Modeler. This nugget stores the trained CHAID model and can be applied to new data for scoring and prediction.

The model nugget is visible both on the canvas (connected at the bottom of the stream) and in the Models panel on the right-hand side of the SPSS Modeler interface (labeled HAS_CHU...). This confirms that the model was successfully built and is available for use.

4.6 Table Node — Output Verification

The Table output node is connected after the Select (Data Known) node and is used to verify the filtered data before it enters the modeling pipeline. It displays the filtered records in a tabular view showing all 23 fields.

The Table node output confirms that the filter condition (DATA_KNOWN = "yes") was applied correctly and that the data contains the expected columns and values before being processed by the Type and CHAID nodes.

Table (23 fields, 31,769 records) #1

File Edit Generate

Table Annotations

	CUSTOMER...	DATA_KNO...	GEND...	AGE	TARIFF	DROPPED_CA...	HANDS...	PEAK_MI...	PEAK_RA...	BILL_PEAK	OFFPEAK_MI...	OFFPEAK_RA...	BILL_OFFPEAK	GADGET...	GADGET...	GADGET...	GADGET...	GADGET_A_REVEN...	GA
1	K100010	yes	Male	4...	CAT 50	1.000	SOP10	36.131	15.000	5.420	7.973	5.000	0.399	F	F	T	F	0.000	
2	K100020	yes	Male	2...	CAT 50	0.000	SOP10	39.437	15.000	5.916	21.153	5.000	1.058	T	F	F	F	11.000	
3	K100030	yes	Male	3...	CAT 50	2.000	SOP20	72.600	15.000	10.890	27.600	5.000	1.380	T	T	F	T	8.000	
4	K100040	yes	Male	2...	CAT 50	2.000	SOP10	72.600	15.000	10.890	27.600	5.000	1.380	T	T	F	T	13.000	
5	K100050	yes	Male	4...	CAT 50	0.000	SOP10	40.608	15.000	6.091	18.824	5.000	0.941	F	F	F	F	0.000	
6	K100060	yes	Male	2...	CAT 50	1.000	SOP10	46.260	15.000	6.939	11.085	5.000	0.554	F	F	F	F	0.000	
7	K100070	yes	Male	3...	CAT 50	1.000	SOP20	56.370	15.000	8.456	4.012	5.000	0.201	F	F	T	F	0.000	
8	K100080	yes	Male	2...	CAT 50	0.000	SOP20	51.043	15.000	14.4[illegible]	34.802	5.000	1.740	F	F	F	F	0.000	
9	K100090	yes	Male	3...	CAT 50	1.000	SOP20	99.000	15.000	14.850	53.100	5.000	2.655	F	F	F	F	0.000	
10	K100100	yes	Male	4...	CAT 50	0.000	SOP20	65.400	15.000	9.810	29.400	5.000	1.470	T	T	F	T	7.000	
11	K100110	yes	Male	3...	CAT 50	2.000	SOP20	65.400	15.000	9.810	29.400	5.000	1.470	F	F	F	T	0.000	
12	K100120	yes	Male	3...	CAT 50	0.000	SOP20	68.123	15.000	10.218	33.880	5.000	1.694	F	F	F	F	0.000	
13	K100130	yes	Male	4...	CAT 50	2.000	SOP20	63.352	15.000	9.503	26.803	5.000	1.340	F	T	T	F	0.000	
14	K100140	yes	Male	2...	CAT 50	1.000	SOP20	41.447	15.000	6.217	16.133	5.000	0.807	F	F	F	F	0.000	
15	K100150	yes	Male	3...	CAT 50	2.000	SOP20	96.000	15.000	14.400	34.800	5.000	1.740	F	T	T	T	0.000	
16	K100160	yes	Male	2...	CAT 50	9.000	SOP20	41.342	15.000	6.201	23.841	5.000	1.192	F	F	F	F	0.000	
17	K100170	yes	Male	2...	CAT 50	1.000	SOP10	100.200	15.000	15.030	31.500	5.000	1.575	F	F	F	F	0.000	
18	K100180	yes	Male	3...	CAT 50	1.000	SOP10	41.918	15.000	6.288	21.827	5.000	1.091	F	F	F	F	0.000	
19	K100190	yes	Male	3...	CAT 50	2.000	SOP10	58.782	15.000	8.817	28.749	5.000	1.437	F	F	F	F	0.000	
20	K100200	yes	Male	4...	CAT 50	0.000	SOP10	84.600	15.000	12.690	27.900	5.000	1.395	F	F	F	F	0.000	
21	K100210	yes	Male	3...	CAT 50	2.000	SOP20	84.600	15.000	12.690	27.900	5.000	1.395	F	F	F	F	0.000	
22	K100220	yes	Male	2...	CAT 50	9.000	SOP20	41.712	15.000	6.257	15.279	5.000	0.764	F	F	F	F	0.000	
23	K100230	yes	Male	3...	CAT 50	2.000	SOP10	71.310	15.000	10.697	38.428	5.000	1.921	F	F	F	F	0.000	
24	K100240	yes	Male	3...	CAT 50	0.000	SOP10	138.600	15.000	20.790	0.000	5.000	0.000	F	F	F	F	0.000	
25	K100250	yes	Male	2...	CAT 50	0.000	SOP20	71.400	15.000	10.710	27.900	5.000	1.395	F	F	F	F	0.000	
26	K100260	yes	Male	4...	CAT 50	1.000	SOP10	33.458	15.000	5.019	22.224	5.000	1.111	F	F	F	F	0.000	
27	K100270	yes	Male	4...	CAT 50	2.000	SOP20	41.990	15.000	6.298	14.465	5.000	0.723	T	T	F	F	15.000	
28	K100280	yes	Male	4...	CAT 50	0.000	SOP10	80.400	15.000	12.060	25.200	5.000	1.260	F	F	F	F	0.000	
29	K100290	yes	Male	2...	CAT 50	0.000	SOP20	90.000	15.000	13.500	45.600	5.000	2.280	F	T	F	F	0.000	
30	K100300	yes	Male	3...	CAT 50	8.000	SOP20	87.600	15.000	13.140	30.300	5.000	1.515	F	F	F	F	0.000	
31	K100310	yes	Male	3...	CAT 50	11.000	SOP20	36.357	15.000	5.454	17.844	5.000	0.892	T	F	T	T	9.000	
32	K100320	yes	Male	3...	CAT 50	1.000	SOP10	40.081	15.000	6.012	8.761	5.000	0.438	T	T	F	T	7.000	
33	K100330	yes	Male	2...	CAT 50	1.000	SOP10	73.549	15.000	11.032	25.030	5.000	1.251	F	F	T	F	0.000	
34	K100340	yes	Male	5...	CAT 50	0.000	SOP20	87.600	15.000	13.140	30.300	5.000	1.515	F	F	F	F	0.000	
35	K100350	yes	Male	3...	CAT 50	2.000	SOP10	61.909	15.000	9.286	13.169	5.000	0.658	F	F	F	F	0.000	
36	K100360	yes	Male	3...	CAT 50	6.000	SOP10	51.385	15.000	7.708	17.483	5.000	0.874	F	T	T	T	0.000	
37	K100370	yes	Male	4...	CAT 50	0.000	SOP20	91.200	15.000	13.680	48.600	5.000	2.430	F	T	F	F	0.000	
38	K100380	yes	Male	3...	CAT 50	2.000	SOP20	66.000	15.000	9.900	36.000	5.000	1.800	F	T	F	T	0.000	
39	K100390	yes	Male	3...	CAT 50	1.000	SOP10	55.800	15.000	8.370	18.000	5.000	0.900	F	T	T	F	0.000	
40	K100400	yes	Male	5...	CAT 50	0.000	SOP20	86.400	15.000	12.960	47.700	5.000	2.385	F	F	F	F	0.000	
41	K100410	yes	Male	4...	CAT 50	7.000	SOP20	86.400	15.000	12.960	47.700	5.000	2.385	F	F	F	F	0.000	
42	K100420	yes	Male	3...	CAT 50	1.000	SOP10	43.200	15.000	6.480	0.000	5.000	0.000	F	F	F	F	0.000	

OK

6. Workflow Execution Steps

Step 1: Import Dataset

An Excel source node was added to the canvas and configured to point to the telecom customer dataset file. The node loads all 31,769 records with 23 fields into the stream for processing.

Step 2: Apply Data Filter (Select Node)

A Select node was connected to the Excel source and configured with the condition DATA_KNOWN = "yes" in Include mode. This filters the dataset to retain only records with known churn outcomes, making it suitable for supervised model training.

Step 3: Verify Data (Table Node)

A Table output node was connected to the Select node to preview the filtered output. This step confirms that the filter worked correctly and that the data shows 10 records with all 23 fields intact. This is an important data quality check before proceeding to modeling.

Step 4: Define Field Types (Type Node)

A Type node was connected to the Select node. The "Read Values" function was used to scan all fields and automatically detect measurement levels. The roles were then manually reviewed and configured: CUSTOMER_ID and DATA_KNOWN were set to "None", HAS_CHURNED was set as the Target, and all remaining relevant fields were set as Input predictors.

Step 5: Build CHAID Model

A CHAID modeling node was connected to the Type node with HAS_CHURNED selected as the target field. Running this node trains the decision tree on the filtered, typed dataset and produces a Model Nugget. The nugget captures all the learned decision rules and thresholds.

Step 6: Apply Model & Score

The generated CHAID Model Nugget was connected in the stream and applied to data to produce predictions. The nugget adds new fields to the dataset including the predicted class (HAS_CHURNED predictions) and associated confidence scores.

7. CHAID Model — Key Concepts

7.1 How CHAID Works

CHAID builds a decision tree by recursively splitting the data at each node:

- It evaluates all available input predictors at every split point.
- For each predictor, it tests whether the split is statistically significant using the Chi-square test.
- The predictor with the most significant split (lowest p-value) is chosen for the split.
- The process continues until stopping criteria are met (minimum node size, maximum tree depth, or no significant splits remaining).

7.2 Interpreting Results

The CHAID model segments customers into groups based on their likelihood to churn. Each terminal node (leaf) in the decision tree represents a customer segment with a specific churn probability. Business users can use these segments to:

- Identify high-churn-risk customer profiles.
- Design targeted retention campaigns for at-risk segments.
- Understand which factors (e.g., age, tariff, dropped calls) most influence churn.
- Score new customers without known churn status to predict their likelihood of leaving.

8. Summary & Conclusion

This project successfully demonstrates the end-to-end implementation of a customer churn prediction model using IBM SPSS Modeler, covering all key stages of the CRISP-DM (Cross Industry Standard Process for Data Mining) framework, including business understanding, data understanding, data preparation, modeling, evaluation, and deployment. Each phase was carefully executed to ensure that the dataset was cleaned, transformed, and structured appropriately for analysis, resulting in improved data quality and model performance.

The use of the CHAID (Chi-square Automatic Interaction Detector) decision tree algorithm plays a crucial role in this project, as it effectively segments customers based on significant predictor variables such as usage patterns, billing information, and service preferences. The model identifies relationships between independent variables and churn behavior, making it easy to interpret and visualize. One of the key strengths of CHAID is its ability to handle categorical data and create meaningful splits, which helps in generating clear and actionable customer groups.

The results of the model provide valuable insights into customer behavior by highlighting high-risk churn segments. These insights enable telecom companies to take proactive measures to retain customers by designing targeted retention strategies. For example, customers identified as high-risk can be offered personalized discounts, better service plans, loyalty benefits, or improved customer support. Additionally, medium-risk customers can be monitored and engaged through promotional offers to prevent them from moving toward churn.

Overall, this project not only demonstrates technical implementation but also emphasizes practical business application. The predictive model serves as a decision-support tool for management, helping organizations reduce customer attrition, improve customer satisfaction, and increase long-term profitability. By leveraging data-driven strategies, telecom companies can gain a competitive advantage and build stronger customer relationships.